Cultivar HW8

Gilbert Fellingham

10/13/2026

We will continue with our examination of the cultivar data, this time focusing on variance components. The models we have examined thus far have all had two variance components, σ^2_{error} and σ^2_{field}. It is not unreasonable to think that there may be other variance components of interest. I look at interactions between fixed and random effects as random, and so we might consider $\sigma^2_{field\times cultivar}$ and $\sigma^2_{field\times inoculant}$. In this assignment we will focus on finding a model with appropriate variance components included. Use your cell means model for the fixed effects, and keep the fixed effects the same in all the models you will consider in this assignment. I'm assuming you will use *lmer* for all the models using the frequentist paradigm, and when I want a Bayesian analysis I will ask for it explicitly.

1. Redo the model we have been using with only variance components for σ^2_{error} and σ^2_{field}. What are the point estimates for these two quantities?
2. What are the AIC and BIC for this model?
3. Now use Nimble to get estimates for the variances. Use gamma priors with shape=1.1 and rate=1 for both variance components. What are the point estimates for the two variance components?
4. What is the WAIC for this model?
5. Now add a variance component for $\sigma^2_{field\times cultivar}$. This may take adding a column in the data set. What are the point estimates for the three variance components?
6. What are the AIC and BIC for this model?
7. Using a likelihood ratio test and the ML models, which model is better?
8. Now use Nimble to get estimates for the model in number 5. What are the point estimates for the three variance components?
9. What is the WAIC for this model?
10. To the model with all three variance components, add $\sigma^2_{field\times inoculant}$. What happens?
11. To the model with only σ^2_{error} and σ^2_{field}, add $\sigma^2_{field\times inoculant}$. What happens?
12. Now use Nimble to get estimates for the model in number 9. What are the point estimates for the three variance components?
13. What is the WAIC for this model?
14. What happens to the convergence diagnostics for the model in number 10?